

Presentation Comparison Report

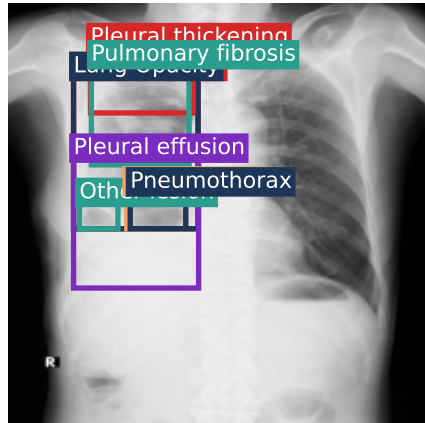
Dense pleural and airspace multi-finding case Case: ba5e3409250a85483d6e39be759bc102

Review status: needs_human_review Confidence: moderate

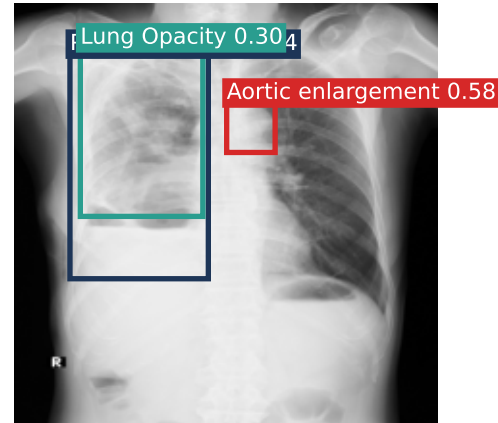
Original CXR



Ground-truth boxes



YOLO support overlay



Ground Truth

- Consolidation
- Lung Opacity
- Other lesion
- Pleural effusion
- Pleural thickening
- Pneumothorax
- Pulmonary fibrosis

Swin Classification

Predicted: Consolidation, Lung Opacity, Pleural effusion, Pleural thickening, Pulmonary fibrosis
Correct: Consolidation, Lung Opacity, Pleural effusion, Pleural thickening, Pulmonary fibrosis
Missed: Other lesion, Pneumothorax
Overcalled: None

YOLO Labels

Predicted: Aortic enlargement, Lung Opacity, Pleural effusion
Correct: Lung Opacity, Pleural effusion
Missed: Consolidation, Other lesion, Pleural thickening, Pneumothorax, Pulmonary fibrosis
Overcalled: Aortic enlargement

Agentic Workflow Summary

Supported findings: Pleural effusion, Lung Opacity, Consolidation, Pleural thickening, Pulmonary fibrosis

Uncertain findings: Aortic enlargement

Final report labels: Consolidation, Lung Opacity, Pleural effusion, Pleural thickening, Pulmonary fibrosis

Pattern buckets: Pleural abnormality, Airspace or infectious-inflammatory pattern, Chronic interstitial or fibrotic pattern, Possible urgent thoracic abnormality

Evidence and Report Summary

Findings

The chest X-ray demonstrates multiple abnormalities. There is evidence of pleural effusion with localization support from both models. Lung opacity and consolidation are present, suggesting an airspace process. Pleural thickening is also identified. Additionally, there are findings consistent with pulmonary fibrosis, indicating chronic interstitial changes. Aortic enlargement is detected by the localization model but receives lower confidence from the classification model, creating uncertainty about this finding.

Impression

Multiple thoracic abnormalities are present including pleural effusion, lung opacity with consolidation, pleural thickening, and pulmonary fibrosis. The combination of acute airspace disease and chronic fibrotic changes suggests a complex pulmonary process. Possible aortic enlargement requires further evaluation. The presence of pleural effusion and consolidation may represent urgent findings requiring clinical correlation.

Top Swin Probabilities

- Pleural effusion: 0.967
- Lung Opacity: 0.933
- Consolidation: 0.862
- Pleural thickening: 0.848
- Pulmonary fibrosis: 0.827
- Atelectasis: 0.443

Presentation Takeaway

Swin correct labels: 5 / 7
YOLO correct labels: 2 / 7
Key miss for Swin: Other lesion, Pneumothorax
Key miss for YOLO: Consolidation, Other lesion, Pleural thickening

Ground-Truth Boxes Top YOLO Detections

- | | |
|--|---|
| - Consolidation @ [78.0, 89.1, 228.9, 272.7] | - Aortic enlargement: 0.58 @ [256.2, 121.3, 315.0, 180.2] |
| - Lung Opacity @ [78.0, 89.1, 228.9, 272.7] | - Pleural effusion: 0.44 @ [66.3, 62.0, 234.0, 333.3] |
| - Other lesion @ [85.1, 241.8, 131.8, 272.7] | - Lung Opacity: 0.30 @ [78.8, 54.4, 227.3, 257.7] |
| - Other lesion @ [142.4, 227.9, 213.6, 272.7] | |
| - Pleural effusion @ [78.0, 188.1, 229.6, 344.5] | |
| - Pleural thickening @ [98.4, 54.0, 222.2, 132.3] | |
| - Pneumothorax @ [146.7, 230.0, 214.0, 272.5] | |
| - Pulmonary fibrosis @ [199.6, 74.8, 218.0, 195.2] | |

Thresholds Used

Swin confidence cutoff: 0.25
YOLO detection confidence cutoff: 0.25
Aortic enlargement is qualitative, not a fixed numerical threshold

Confidence Calculation

How this case's confidence was calculated

Per-finding band cutoffs: high ≥ 0.80 , moderate ≥ 0.50 , low < 0.50 .

Swin label threshold = 0.50. YOLO detection threshold = 0.25.

Pleural effusion: high from score = $\max(\text{Swin}(0.97), \min(1.00, 0.8 \cdot 0.97 + 0.2 \cdot 0.44)) = 0.97$

Lung Opacity: high from score = $\max(\text{Swin}(0.93), \min(1.00, 0.8 \cdot 0.93 + 0.2 \cdot 0.30)) = 0.93$

Consolidation: high from score = $\text{Swin}(0.86)$

Pleural thickening: high from score = $\text{Swin}(0.85)$

Pulmonary fibrosis: high from score = $\text{Swin}(0.83)$

Aortic enlargement: low from score = $\max(\text{Swin}(0.35), \min(1.00, 0.6 \cdot 0.35 + 0.4 \cdot 0.58)) = 0.44$

Agentic AI report band reference: `moderate` occurred in 58 / 300 cases with exact-match accuracy 15.5%.

Interpretation note: the report-level agentic AI confidence band is constrained to high / moderate / low in the schema, but the model chooses that band. The per-finding confidence lines above are deterministic and computed from the Swin probability, YOLO support, and the label taxonomy.

This is an AI-generated decision-support summary that requires human radiologist review and clinical correlation before any diagnostic c